

IC 182: History of Science and Technology

Course Code: IC 182

Course Name: History of Science and Technology

L-T-P-C: 3-0-0-3

Prerequisites: -

Students intended for: B. Tech. 1st year

Elective or Core: IC Elective

Mutual Exclusion: -

Approval: 59th BoA

Course contents

- **Evolution of modern science:** [7 Lectures]
 - Science in Ancient Greek and Egyptian civilizations (e.g. Pythagoras, Ptolemy etc.), its decadence in the dark ages
 - European scientific revolution: Copernicus, Galileo, Kepler, Brahe, Newton, Maxwell, Planck, Einstein
 - Renaissance and Industrial Revolution
- **Unit 2:** [4 Lectures]
 - Notable figures in evolution of mathematics, the thought processes and their contributions: Descartes, Leibniz, Laplace, Fourier, Gauss, Euler
- **Notable Indian Scientists:** [6 Lectures]
 - SN Bose, JC Bose, CV Raman, MN Saha, Ramanujan
- **Developments in Bio sciences:** [2 Lectures]
 - Darwin, Wallace, Robert Cook, Mendel
- **Ancient science in India:** [8 Lectures]
 - Examples from Indus valley civilization, Other examples about architecture, metallurgy, planning etc.
- **Developments in Indian astronomy and mathematics:** [5 Lectures]
 - Surya Siddhanta, Bhramagupta, Aryabhatta, Bhaskara, Varahamihira
 - Discovery of zero and other related concepts
 - Kerala school of mathematics
- **Overview of Ayurveda:** [5 Lectures]
 - Sushruta samhita, Charaka samhita, Ashtanga Hridaya
- **Unit 8:** [5 Lectures]

– Overview of the Darshanas, and the Indian thought systems

References:

1. D.P. Chattopadhyay, **History of science and technology in ancient india**
2. David Wootton, **The invention of science: A new history of scientific revolution**
3. David Lindberg, **The beginnings of western science**
4. Edward Dolnick, **The clockwork universe: Issac Newton, the royal society, and the birth of the modern world**